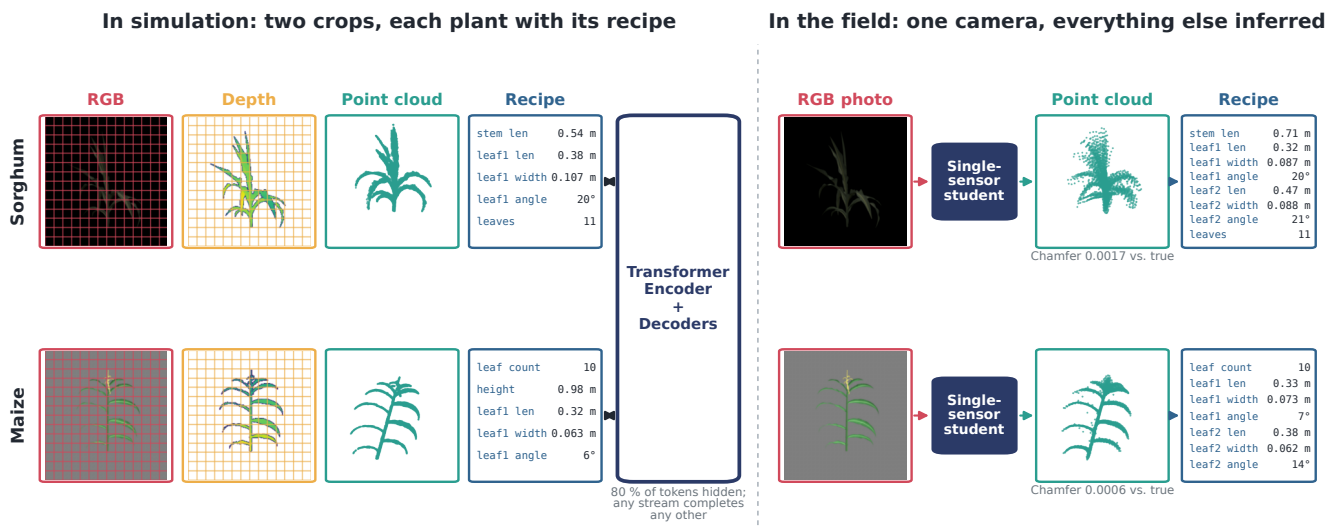

In simulation: two crops, each plant with its recipe
In the field: one camera, everything else inferred
RGB
Depth
Point cloud
Recipe
Sorghum
stem len 0.54 m
leaf1 len 0.38 m
leaf1 width 0.107 m
leaf1 angle 20°
leaves 11
Maize
leaf count 10
height 0.98 m
leaf1 len 0.32 m
leaf1 width 0.063 m
leaf1 angle 6°
Transformer Encoder + Decoders
80 % of tokens hidden; any stream completes any other
RGB photo
Single-sensor student
Point cloud
Recipe
stem len 0.71 m
leaf1 len 0.32 m
leaf1 width 0.087 m
leaf1 angle 20°
leaf2 len 0.47 m
leaf2 width 0.088 m
leaf2 angle 21°
leaves 11
Chamfer 0.0017 vs. true
leaf count 10
leaf1 len 0.33 m
leaf1 width 0.073 m
leaf1 angle 7°
leaf2 len 0.38 m
leaf2 width 0.062 m
leaf2 angle 14°
Chamfer 0.0006 vs. true
held-out Sorghum_10_04 and Maize_1_plant_0004_04 (raw files in paper/data); right: real outputs, sorghum zero-shot (pretrained arm), maize distilled; no ground truth shown